GALGOTIAS COLLEGE OF ENGINEERING & TECHNOLOGY
GREATER NOIDA
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

BTECH (SEM V)

ARTIFICIAL INTELLIGENCE (BCAI-501)

UNIT-3 QUESTION BANK

Section A: Short Answer

1. Differentiate between forward chaining and backward chaining.
2. Apply the concepts of upper ontology with an example.
3. Explain the process of unification with an example where two terms are unified.
4. Illustrate the concept of knowledge representation in artificial intelligence.
5. Define a predicate, and give an example in the context of FOPL.
6. Write a Prolog program to check if a person is an ancestor of another.
7. Explain the role of quantifiers (∀, ∃) in FOPL with examples.
8. Provide an example of unification in Prolog, including cases where unification succeeds and fails.
9. Explain forward chaining with an example of its use in rule-based systems.
10. Write FOPL expressions for the following statements:
    a. All humans are mortal.
    b. Some students like mathematics.

Section B: Long Answer

11. Represent the information used in the mentioned problem in predicate logic. "There is a monkey at the door in a room. In the middle of the room a bunch of bananas is hanging from the ceiling. The monkey is hungry and wants to get the banana but we cannot stretch high enough from the floor. At the window of the room there is a box".
12. Discuss how ontological engineering contributes to developing reasoning systems.
13. Explain the process of defining facts, rules, and queries in Prolog with examples.
14. Prove that following sentence is valid "If price fall then sell increases. If sell increases then Jhon makes the whole money. But John doesn't make the whole money. Therefore, price do not fall".
15. Explain First-Order Predicate Logic and its use in knowledge representation.
16. Differentiate between reasoning with default information and standard reasoning systems?
17. Convert each of the following sentences
    - first into Predicate logic
    - and then in Clausal Form/Prenex Normal Form
    - and then do Resolution for goal: - Scrooge is not a child.

    (i) Every child loves Santa.
    (ii) (ii) Everyone who loves Santa loves any reindeer.
    (iii) (iii)Rudolph is a reindeer, and Rudolph has a red nose.
    (iv) (iv) Anything which has a red nose is weird or is a clown.
    (v) (v) No reindeer is a clown.